UNIT 1

Exercise 1: Starting Excel Program

- (To be completed in the Excel program.)

Start the Excel program. Move your mouse pointer to the ***FILE*** option on the menu bar and click to see its pull-down options. Do not click any pull-down options yet. You can escape or cancel from a pull-down list by either pressing the [ESC] key on the keyboard or clicking outside of the display list. Do one of these to remove the ***FILE*** pull-down list.

Exercise 2: Toolbars

Switch to the Excel program. Put your mouse pointer over each button on both of the toolbars and view its *ToolTip*. You will see some buttons similar to other windows applications, especially if you are using other Microsoft programs like Word. We will use more of these buttons in other lessons.

Exercise 3: Active Cell or Selected Cell

Move the active cell. First, notice where the bordered box is. When you begin working in a new spreadsheet, the active cell will be cell A1. Look at the left most cell on the formula bar as you change the active cell.

Exercise 4: **Marking a block**

Let us say you want to mark the block from B2 to D6. This will encompass 15 cells. To do this, click on cell B2 and hold down the left mouse button; then drag the mouse down and to the right. (Make sure you continue to hold the mouse button down.) If you see that cells are being blackened, you are progressing correctly. Continue to move the mouse arrow down to cell D6 and then release the mouse button. This should result in the desired 15 cells being in a blackened rectangle (the original active cell of B2 is not blackened but is included in the block).

Using the mouse is popular when marking a block, but you can also hold down the [Shift] key and simultaneously press arrow keys. Do not release the [Shift] key until all the cells you want are highlighted. Try this method of blocking as well.

Exercise 5: **Data Entry**

Enter this information in the same cells as indicated here.

	A	B	C	D	E	F	G
1		Red	Blue	Green	Totals		
2	Mary	23	25	25			
3	Sam	16	18	19			
4	Bill	19	33	4			
5	Sue	24	22	33			
6	Mandy	17	27	21			
7	Tommy	9	5	18			
8	Totals						
9							
10							

Note: Text will display to the left and numbers to the right in each cell. Leave the justification alone. In Lesson 2 you will learn about justification options.

Exercise 6: **Entering Formula**
First make cell B8 your active cell. Enter the simple formula: =23+16+19+24+17+9 into B8. (Be sure to use number 1 and not the letter l and 0 and not the letter o.) Remember to begin this formula with the = sign. You should see the result of your formula 108 displaying in that cell in the spreadsheet. Look at the formula bar; it will be displaying what you typed as the formula.
Move to cell C8. Enter the formula: =C2+C3+C4+C5+C6+C7 into cell C8. Again compare the cell results displaying in that cell with the formula bar at the top of the screen.

Having entered the two formulas and having the totals appear, now enter a different number in both column B2 and C2 for Mary and see what happens to the total results in cell B8 and C8. What happened? For column B there was no correcting update to the total. This was because of the inflexibility of the formula, whereas in column C the formula referenced the position and was correctly updated when the numbers above changed. This is important to understand.

Exercise 7
Try entering this formula in D8. The formula should be: =SUM(D2:D7). Note the results. Again, enter a new number for Mary in D2 and see what the results are. If your formula did not function properly, check that the formula has been correctly entered. (The : between the two cells tells Excel to treat this as a block of cells, from D2 through D7.)

Exercise 8

Move to cell B8 and press the [DEL] key to erase this formula. Click on  AutoSum button near the middle of the Standard Toolbar. The symbol is the Greek sigma, and looks like a fancy E. AutoSum is designed to look for numbers above it and add those numbers together; if there are no numbers above, it looks for numbers to the left of the active cell. In this example, there are numbers above cell B8, so Excel displays a sum function of =SUM(B2:B7). Press the [ENTER] key to accept this formula. There is a new total in cell B8.

The AutoSum can be applied to several columns or several rows at once. For example, use your mouse to block from cells E2 through E8. Now click on the AutoSum and the totals for each row will be placed in column E.

Exercise 9: Printing the Workbook
Click on ***File*** on the menu bar and then ***Print***. You have options to select what or how much you want to print under the section of "Print What" in the Print dialog box. The default is to print the active sheet.

Before you print, click on the [Print Preview] button in the Print dialog box to see what it will look like. The small file you have created will not fill up the sheet of paper. You may

want to click on the button [Margins] in the Print Preview screen to display lines for the location of top, bottom, and side margins. When you put the mouse over a dotted margin line, it changes shape from a magnifying glass to a cross arrow. Click and drag on one the dotted margin lines to move it in the display. You can use this technique to lower the display on the paper and move it to the middle. Go ahead and click on the [Print] button in the Print Preview screen to print the file to your printer.

Unit 1 Exercises

1. Why is an Excel file called a workbook?

2. What are constant values in an Excel file? Give some examples.

3. Bring up a blank workbook in Excel and press the [END] key, then a right arrow key. Where does the cursor move to? Again, press the [END] key, then the down arrow key. Where is the active cell sitting? What is the address of the lower right cell on a workbook page?

   Open up the file `FUNDRAS.XLS` that you saved and use these same [END] and arrow keys in this file. What happens?

   Practice using these movement keys in the file:

   [PageUp]
   [PageDown]
   [End], then [Home]
   [Home]
   [Ctrl]+[Home] (pressing at the same time)

4. Using the **`FUNDRAS.XLS`** file, click on ***File*** on the menu bar, then ***Save As***. Change the filename to **`PRACTICE.XLS`**
5. There are two ways to print a file, by clicking on the "Print" button on the Standard toolbar and by selecting ***File*** then ***Print***. What is the difference in using these two options? Try both with your `PRACTICE.XLS` file.

6. Which of these are valid cell addresses?

7.	G3000	DH30	20AD
8.	b20	Ab6000	550

9. Using the `PRACTICE.XLS` file, highlight several cells and press the [DELETE] key. Now click on the "Undo" button on the Standard toolbar. What happens?

10. Close any spreadsheet files that are open. Click on ***File***, then ***New*** to bring up a blank layout. Using the same skills you used earlier in this lesson, create this spreadsheet:

	A	B	C	D	E	F
1	Attendance					
2	Class	Fall	Winter	Spring	Summer	Totals
3	Plants	40	50	10	30	
4	Insects	100	50	60	35	
5	Money	10	40	60	30	
6	Leaders	35	75	10	40	
7	Totals					

Use the "AutoSum" button to calculate the totals. Save this file under the name of CLASSES. Practice printing this file; you can use the Print Preview screen and adjust the margins with the mouse first, if you want.

11. Here is another application to create:

	A	B	C	D	E
1	Meeting	Registration			
2			Brunch	Dinner	Tour
3	Mary	Jones	1	2	1
4	Bill	Ramon	1	1	1
5	Sue	Black	2	3	3
6	Angie	Rivers	1	1	2
7	Harry	Adams	1	1	1
8	Totals				

Save this under the name of REGISER.

Take the time to go through some of the online lessons found under HELP, then Examples and Demos. Focus for now on the lessons under "Selecting Cells," "Using Toolbars," and "Entering Data."

12. Use the "online tips" database to read more about various items covered in this lesson. Click on HELP, then "Search for Help on" and type in these phrases.

```
deleting, cells
AutoSum button
Print area
```

When Excel finds the term in the alphabetical listing, click on the "Show Topics" button and then pick out a topic from the lower half of the help screen. Click on the button "Go To" to read the online help. From the Help window, you can click on the "Print" button

on the menu bar to have a printed copy, if you want. Close the Help window when you are through.

UNIT 2

Exercise 1: Changing Column Widths

Open the **FUNDRAS.XLS** file created in the first lesson. Use ***File***, ***Save As*** and save the file under a new name of **PRACTICE.XLS** (so you have a clean version of FUNDRAS.XLS later). Position your cursor on cell A2, double-click on the cell and edit the name so you have a first and last name in each cell. Work your way down the column, editing each name, so the names are:
Mary Anderson
Sam Jones
Bill Wrightstown
Susey Olvidad
Mandy Dorson
Robert Hinz

You can see right away that column A is not wide enough. You cannot read the complete names. Make B4 the active cell (where the number 19 is) and press the [DELETE] key to erase the number. When the cell or cells next to a wide text entry are empty, Excel will often display the complete cell's contents. But if there is information in a cell adjacent to a wide text entry, the text display is truncated. Type the number **19** again in cell B4.
Click on the **A** in the frame area to select the entire column. Click on ***Format*** on the menu bar, then ***Column***, then ***AutoFit Selection*** to widen column A.
Use the other two techniques to narrow B, C, D, and E.

After you make column B narrower, change at least one number in column B to **100000**. This large number in a narrow column illustrates how Excel deals with a column not large enough for a value--Excel displays pound symbols in the cell (#####). This happens most often on cells with formulas or functions, but can also happen on cells with dates, since they are also values. If pound symbols appear in a cell, simply widen a column so that the entire number will display.

Exercise 2: Inserting a column or row
Click on **B** in the frame area and then click on ***Insert***, then ***Columns***. Column B and the other columns to its right are moved to the right. Type in a unit (or department) name in column B for each of the "fundraisers."
As a further example, click on row 1 in the frame area so the entire row is highlighted. Click on ***Insert***, then ***Rows*** for a total of three times, so you have three blank rows at the top of the spreadsheet.

Be sure to click on the  "Save" button on the Standard toolbar to save a current copy of the **PRACTICE.XLS** file.

Exercise 3: **Deleting a column or row**
Select the entire row at the top of the PRACTICE file and then click on ***Edit***, then ***Delete*** to remove this row.

Exercise 4

Make these changes in the **PRACTICE** file:

1. Have a new column D be Black and type in numbers in that column for each person. Use the Σ "AutoSum" button to total that column in cell D10.
2. Have a new column G be Purple and type in numbers in that column for each person. Again use the Σ "AutoSum" button to total that column in cell G10.
3. Highlight the totals in H and press the [DELETE] key to erase the numbers. Highlight from cells H4 through H10 and click on Σ "AutoSum" to calculate new totals in the last column.
4. Move to cell A1 and type in the heading of **FUND RAISING PROJECT**.

Be sure to save this new version of **PRACTICE.XLS**.

The first three rows of the spreadsheet should now look like this:

	A	B	C	D	E	F	G	H
1	Fund Raising Project							
2								
3		Unit	Red	Black	Blue	Green	Purple	Totals

In the same way that insertions can sometimes make calculations inaccurate, deleting certain columns or rows can also interrupt a formula and the formula may have to be adjusted or re-entered.

Exercise 5: Formatting
Highlight the Totals column from cell H3 through H10 and click on the **B** "Bold" button on the toolbar to apply boldface. Highlight from B3 through H3 for the column headings and make them italic by clicking on the *I* "Italic" button. If you position your cursor on cell H3 you will see that the "Bold" and "Italic" buttons are depressed since the features are turned on here. You can turn them off by clicking on the buttons again.

Exercise 6
Make the active cell A1 in the PRACTICE file. Change it so it is bold and a 14 point size, using the 10 "Font Size" button and the **B** "Bold" button on the Formatting toolbar. Highlight cells A1 through H1 and select "Center Across Columns". Click where the heading now appears and check the formula bar. Although the heading appears to be in the middle, the formula bar shows no text. Click on A1 and look at the formula bar. If you wanted to change the text, you would have to go to cell A1 to edit the cell.

Highlight cells C3 through H3 and right align this text, so the headings line up with the numbers.

Save this version of PRACTICE.XLS.

Exercise 7: Copying and moving

You are going to add three rows at the bottom for showing the maximum value, minimum value, and average value for each color (in each column). In cell A11 type the heading **Largest Number**. Place the cursor on cell C11 and type in this function: **=MAX(C4:C9)** . Rather than have to type this formula in each cell across the row, you can copy it. The active cell is still C11, where the formula is located. If you look at the cell, you will see a small dot in the lower right corner; this dot is called the *AutoFill handle*. Put your mouse cursor on the AutoFill handle and when it becomes a simple plus sign, click on and drag the handle to the right through cell H11. When you release the handle, information from the "source" cell is copied into the other cells.

The cells in row 11 will display as values, but if you click on these cells you will see that the =MAX function was copied to the right. Check out the formulas in D11, E11, F11, G11, and H11. You will notice that Excel has adjusted the formula as it was copied. That is, references to the column change as the column changes.

In cell A12 type the heading **Smallest Number**. Then in cell C12 type this formula **=MIN(C4:C9)**. The smallest value in that column will display. Again, copy the formula across, by dragging on the AutoFill handle in cell C12.

In cell A13 type the heading **Average**. Then in cell C13 type this formula **=AVERAGE(C4:C9)**. The average for that column displays. Again, copy the formula across, by dragging on the AutoFill handle in cell C13. (These numbers may display decimal values, which you will fix soon.)

Change some of the numbers in your spreadsheet and see if the totals recalculate and new maximum, minimum, and average values appear. Save this copy of the PRACTICE.XLS file.

Exercise 8

Highlight the block A10 through H13 and move this block to cell A18. Click on one of the formula cells and look at the cell ranges in the formula. Have the formulas changed? Use the  "Undo" button on the toolbar to move the range back.

Exercise 9: Using Dates

Go to cell F17 and type in the text: **Data as of**
Now go to cell H17 and type in the date: **10/10/96**
Save this new version of PRACTICE.XLS.

Exercise 10: Using Style Options on the Formatting Toolbar

In the exercises later you will be experimenting with some of these, but for now select cells C13 through H13. Click on the  "Increase Decimal" button and watch what happens. Then click on the  "Decrease Decimal" button until you have only 1 decimal place displaying. Save this version of the PRACTICE file.

Unit 2 Exercises and Questions

1. What are two ways to change a column's width?

2. If you enter data in a spreadsheet in the first row and want to lower it to add a title to the top, what do you do?

3. If you click on the "Bold" button to make a cell boldface, how can you remove the bold trait?

4. What button selection can be used to center a title at the top of a spreadsheet?

5. Make sure you have saved a copy of the PRACTICE.XLS file. Select all of column G (Purple) and click on ***Edit*** and ***Delete*** to delete the column. What happens to the formula in the last column? Is it still accurate? Use the "Undo" button to replace the deleted column and close the window.

6. Take the time to go through some more of the Examples and Demos material under ***Help***. Focus on "Editing a Worksheet" and "Formatting a Worksheet" (there are many items in both of these that we have not covered in this lesson).
7. Use the "online tips" database under ***Help*** to read more about various items. Click on ***Help***, then ***Search for Help on*** and type in these phrases:

```
Inserting rows
Deleting, rows
Formats, removing
AutoFill (then overview of using AutoFill)
Decrease decimal button
Increase decimal button
```

8. Using the CLASSES.XLS file from the first lesson, add in row 8 the average for the class attendance for each season. After typing in the formula in B8, use AutoFill to copy the formula across to F8. Save the file again. Highlight cells B8 through F8 and adjust the decimal precision so only whole numbers are displaying. Experiment with different alignment options for the information in column A and the numbers in columns B through F. Save the file again. Select all the information displaying and change the font to a 14 point size. What happens to the display? You do not need to save this "enlarged" display. Use ***File***, ***Close*** and do not save the changes.

9. Click on ***File***, then ***New***, and on a blank spreadsheet display enter these numbers in a row:
10. 20.50 100.75 30000 40.717 5000.00 0.776

Notice that Excel will drop any trailing zeros even if you type them in. Highlight these cells and experiment with the "Increase Decimal" and "Decrease Decimal" buttons on the Formatting toolbar. Also experiment with the "Comma Style" button and the "Currency Style" button.

11. Using the same file, type in these numbers in another row:
12. 100 200 300

Highlight these three cells and click and drag on the AutoFill handle for several cells across. What happens? This is another example of how you can use the AutoFill feature.

You do not need to save this file.

13. Open up the file `REGISTER.XLS` from the first lesson. Practice making these changes in the file:

- center the heading using the "Center Across Columns" button
- Apply a center justification to columns C, D, and E
- Insert four rows for more people and add in more registrations. You may need to erase the current totals values and redo them with the "AutoSum" button.

It is not always necessary for summary information to appear at the bottom of a display. Especially if you had a long list of registrations, you might want to see the totals quickly without having to scroll to the bottom of the listing.

- Click on row 2 where the column headings are located and insert 5 new rows.
- Delete the total row at the bottom.
- Go to A3 and type in: **`Totals for Brunch`**
- Go to A4 and type in: **`Totals for Dinner`**
- Go to A5 and type in: **`Totals for Tour`**
- In C3 type in: **`=SUM(C8:C50)`**
- In C4 type in: **`=SUM(D8:D50)`**
- In C5 type in: **`=SUM(E8:E50)`**

These three sums would handle additions to the file through row 50. You could make the "bottom" row whatever number you needed to handle your expected registration.

Save this as the new version of REGISTER.XLS.

14. For additional practice, create this spreadsheet. You will need to adjust the width of the columns. Apply the "Comma Style" to the numbers. Adjust the alignment to suit your taste.

	A	B	C
1	Topic	Enrollment	Date
2	Plant Genetics	1150	3/20/96
3	Marketing	400	4/5/96
4	World Wide Web	1000	3/28/96
5	Animal Disease	600	5/5/96

Calculate a total for column B using "AutoSum".

Save this file under the name of **PROJECTS.XLS**.

15. Here is another spreadsheet that will give you practice in creating different types of formulas:

	A	B	C	D
1	Remaining Leave			
2				
3	Balance from previous year			45.0
4	Amount earned per month			15.0
5				
6		Hours	Hours	Annual
7	Month	Earned	Used	Leave Balance
8	Balance - June 95			=D3
9	July	=D4	10.5	=D8+B9-C9
10	August	=D4	5.0	(copy from above)
	(Continue for each month, through June			

Note: this spreadsheet illustrates the simplest type of formula--a reference to a single cell. The values in D3 and D4 would be entered by you, as well as the values in column C. The formula from D9 would be copied down for the entire year, and the formula in B9 would continue to be =D4 all the way down.

After you create this spreadsheet, format the numbers using the "Decimal" buttons. You will also have to manually adjust column A's width, perhaps.

Save this file under the name of **LEAVE.XLS**. After saving the file, change the values in D3 and D4 and in column C and see if you get correct calculations.

Incidentally, this same type of format (and formulas) could be used for doing a checkbook application, with columns for income and expenditures. You would also have a date and perhaps a code for categorizing expenses.

If you created an original spreadsheet in a previous lesson, add formatting changes, like column widths, justification, and number formatting, to your spreadsheet.